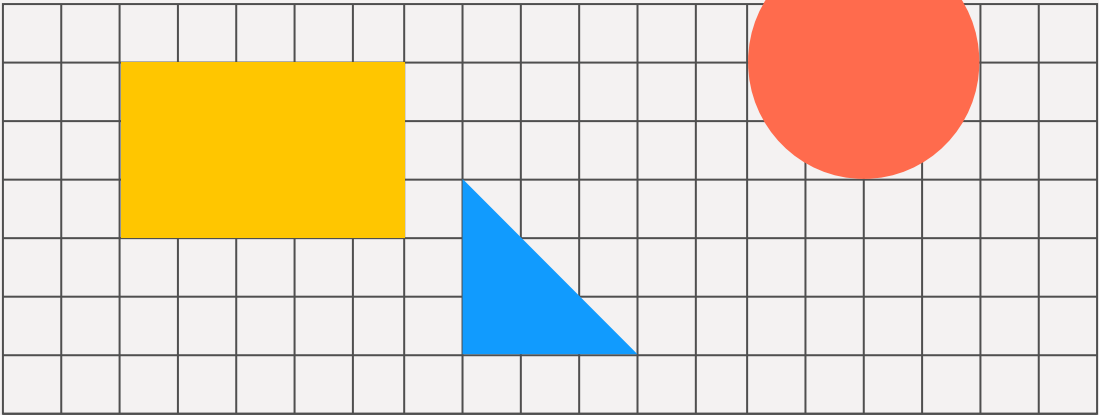


Deep Q Network

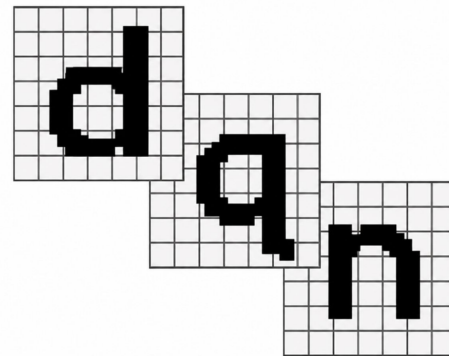
Ember Lu, CMPM 118, Spring 2026



What is DQN?

A Deep Q Network is a deep neural network that uses elements of the Q-Learning algorithm and a memory replay for a faster, more efficient version of reinforcement learning for visual inputs.

This paper discusses the creation of a DQN tested on six Atari games with discrete possible actions (i.e. moving left vs. right) and measuring reward through the course of gameplay.



dqn

Why DQN?

DQN

Determines the reward based on not just a single state of the simulation but a series of states of the simulation combined into one

Pulls from a memory replay so it continuously learns from random previous iterations

RL

Large delay between calculating an action and its reward in RL = many simulation timesteps

Traditional deep learning is typically used for independent data samples and a fixed data distribution while RL is typically used for series of “highly correlated states” and a changing data distribution.

DQN utilizes deep learning and the RL Q-Learning algorithm to make a neural network more efficient and fit for discrete actions.

The Algorithm

Paraphrased from Algorithm 1 in
<https://arxiv.org/pdf/1312.5602>

1. Initialize Replay Memory
2. Initialize Action-Value Function Q with Random Weights
3. For each episode:
 - a. Observe the current state of the simulation (and previous states)
 - b. For each timestep:
 - i. Select and execute a random action or the best action as determined by the Q function
 - ii. Observe the reward and state of the simulation
 - iii. Add the action to the current state of the simulation fed into the algorithm
 - iv. Look back at the memory replay (bank of previous actions) and learn from random samples of memories
 - v. Gradient Descent

Let's Play Atari

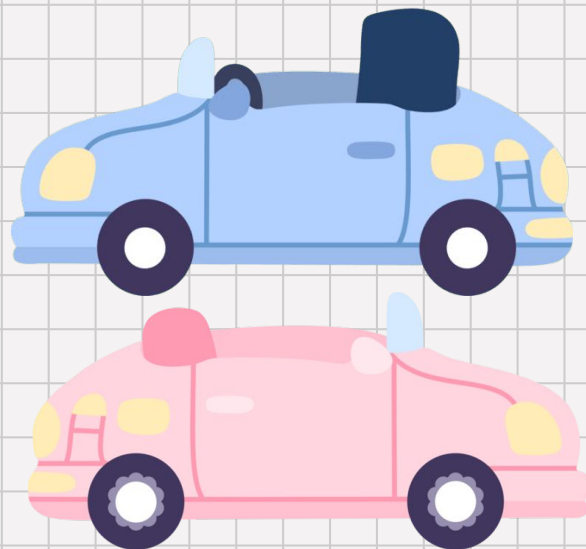
The Results

- **Rewards of 0, 1, and -1 for points scored and actions deducted.**
- **states of the simulation sent through the algorithm were at intervals rather than every frame of the simulation sent in, increasing speed.**
- **Better performance than experts on three Atari games: Breakout, Enduro & Pong**
- **Close to human performance on Beam Rider**
- **Q*bert, Seaquest, and Space Invaders did not meet human performance**



Applying to AVs

- DQN also uses visual input like AVs to determine a main controller's next move based on discrete actions available.
- Since the car is moving through a series of frames of the simulation, having a visual input into DQN that contains information about several previous frames layered is well suited for our purposes
- DQN learns from previous car movements randomly sampled, as stored by the memory replay, not just the car's previous state, which is well suited for modeling car simulations.



Thank You!

[1] V. Mnih *et al.*, “Playing Atari with Deep Reinforcement Learning,” *arXiv.org*, Dec. 19, 2013. <https://arxiv.org/abs/1312.5602>

